DATASCI 347 Final Project Documentation

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Introduction

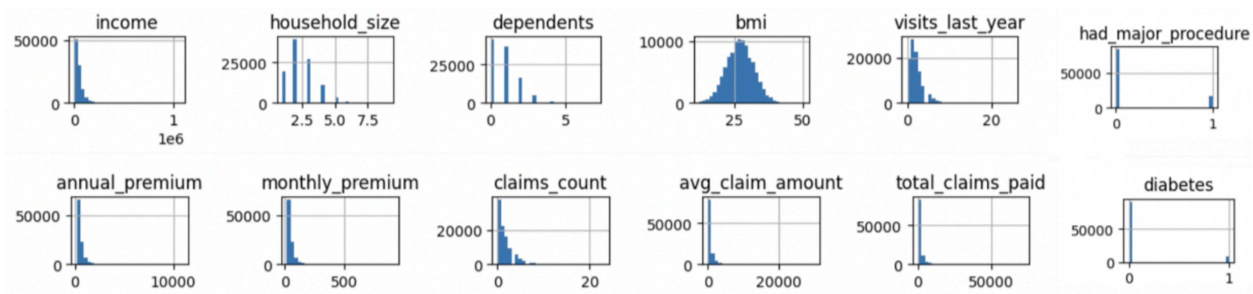
Our project aims to analyze a large problem within the US healthcare industry. In 2024, average US healthcare spending was $14,775 per person, making it the highest in the world. Medical costs come with lots of variation between patients, facilities, and insurance plans making treatment prices unpredictable. As such, it becomes very difficult for patients to choose their healthcare plans correctly. This problem affects most American families as they incur medical costs and decide which insurance plan to purchase. Using our predictor, families can plan ahead of time and understand how their demographics, lifestyle, and healthcare utilization influence their projected medical costs, and therefore make decisions about their insurance plans and other finances.

Our project uses patient level data to predict annual medical spending using demographics, lifestyle variables, pre-existing health conditions, and prior healthcare utilization. We compare multiple models to evaluate how well different methods can estimate medical costs and understand what the key drivers of medical spending are so we can provide insights that help individuals make more informed financial decisions.

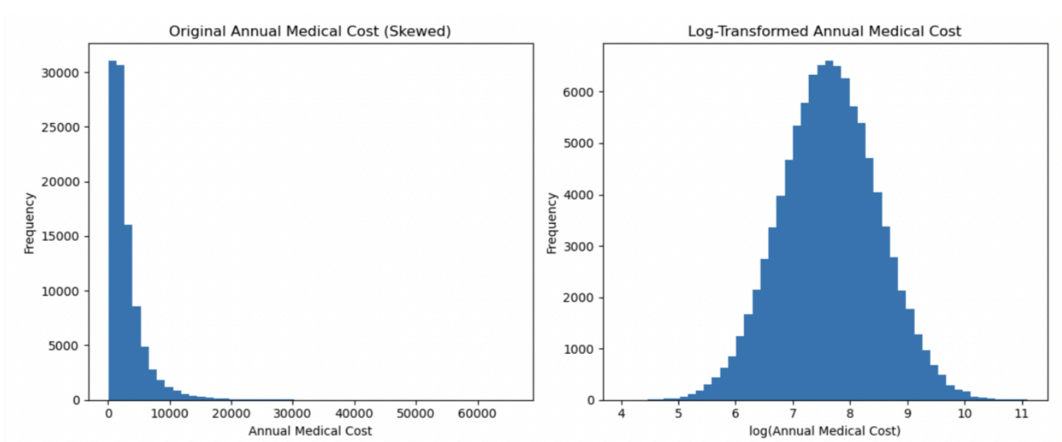
Our approach includes 4 models: linear regression, bagging, random forest, and gradient boosting. Linear regression is our baseline model which will examine if there is a linear relationship between our predictors and our dependent variable medical_cost. It can be useful for identifying the significance of variables and relationships between variables. However, it has many limitations especially since our dataset has variables with non-linear relationships. Our second approach is bagging. Bagging is a good method for non-linear relationships and reduces variance. Since medical costs do not follow a linear pattern, bagging is a good approach for our data. Key components include bootstrapping, parallel training, aggregation hence its name: (B)ootstrap (Agg)regating. Our third approach is Random Forest which is optimal for non-linear relationships. Similar to bagging, it also reduces variance and overfitting, but also decorrelates trees. Random feature selection is used at each split in the tree and it works well with the dataset because of complex relationships and variables and potential variable interactions. Our last approach was Gradient Boosting. Just like bagging and random forest, Gradient Boosting combines multiple decision trees, however, GB trees are dependent and sequential. Each tree builds off the previous to reduce error. It is slow to run (especially with tuning needs) and prone to overfitting.

Setup

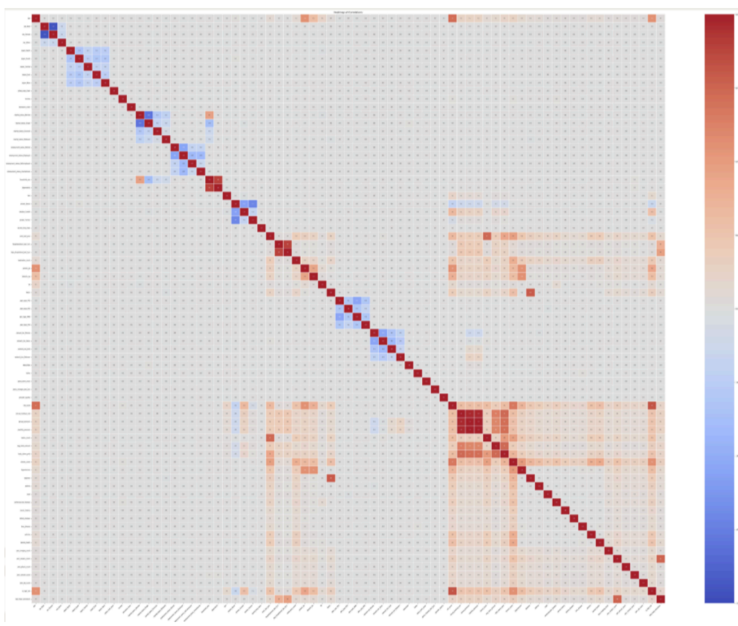
Our dataset comes from Kaggle and is originally sourced from the CDC and National Institute of Health (NIH). It has observations for 100,000 patients across 54 different variables. These variables cover demographics (age, income, bmi, education, sex, region, marital_status), past health conditions (diabetes, asthma, had_major_procedure, cancer_history), lifestyle choices (alcohol_freq, smoker), healthcare utilization (hospitilizations_last_3yrs, medication_count, claims_count), and insurance (avg_claim_amount, monthly_premium, total_claims_paid). Our target variable that we have chosen for our analysis is annual_medical_cost. We aim to predict medical spending for a patient based on their makeup of these different factors.



After performing exploratory data analysis on our dataset we found that most of the variables are right skewed as shown in the figure above. A few notable variables include income and household size. A majority of individuals have incomes above $30,000 and relatively small household sizes at around 3. Our EDA also told us that a majority of patients are unaffected when it comes to prior health conditions, such as diabetes or if they had a major procedure. BMI was also our only normally distributed variable.



When looking at our target variable, annual_medical cost, the distribution is also highly right skewed. Thus, we applied a log transformation to evenly distribute the data to improve model performance and stability. The specific function we applied is as follows: np.log1p(cost)



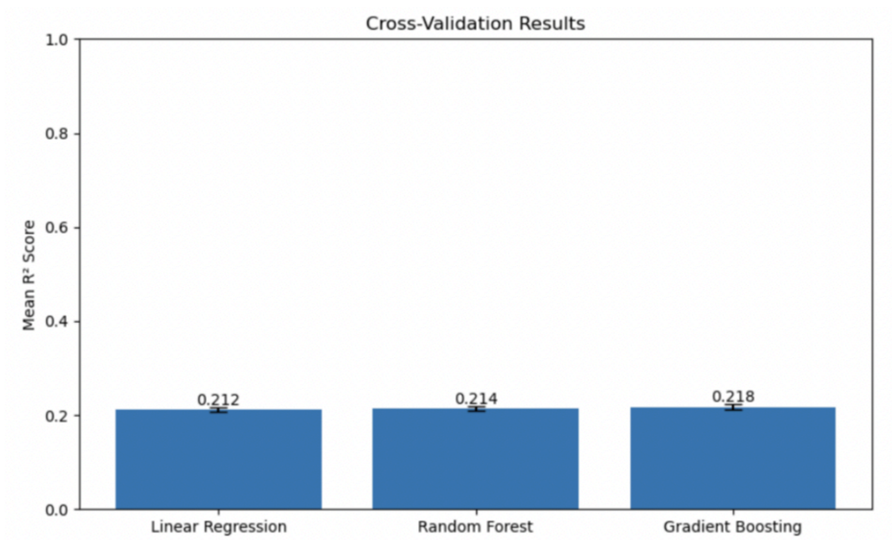
When deciding which features to include in our model we generated a correlation matrix to get an idea of which variables would be the best predictors for annual_medical_cost. Our correlation matrix shows a strong relationship between medical cost and other payment related variables (ie claim amount, monthly premium). Non-payment related variables showed weak linear relations to medical cost. Payment variables, however, are not best suited as predictors as they are too directly related and inaccessible in real-world settings. As a result, we decided to exclude them from our models.

The metrics we used to evaluate our models were R^2 and RMSE. The personal computing was conducted on a Jupyter notebook with Python. Linear Regression was our baseline model to compare to others and did not require any hyperparameters. Bagging works as it is an aggregated prediction from decision trees made by bootstrap samples of data. Our parameters were: Base model: DecisionTreeRegressor, n_estimators=200. Random Forest is like Bagging, however, random subsets of variables are chosen at each split. Our parameters were: n_estimators=200, max_depth=10, min_samples_leaf=5. Lastly, Gradient Boosting involves having each tree sequentially built to reduce error and final prediction is the sum of trees. Our parameters were: n_estimators=200, learning_rate=0.05, max_depth=3.

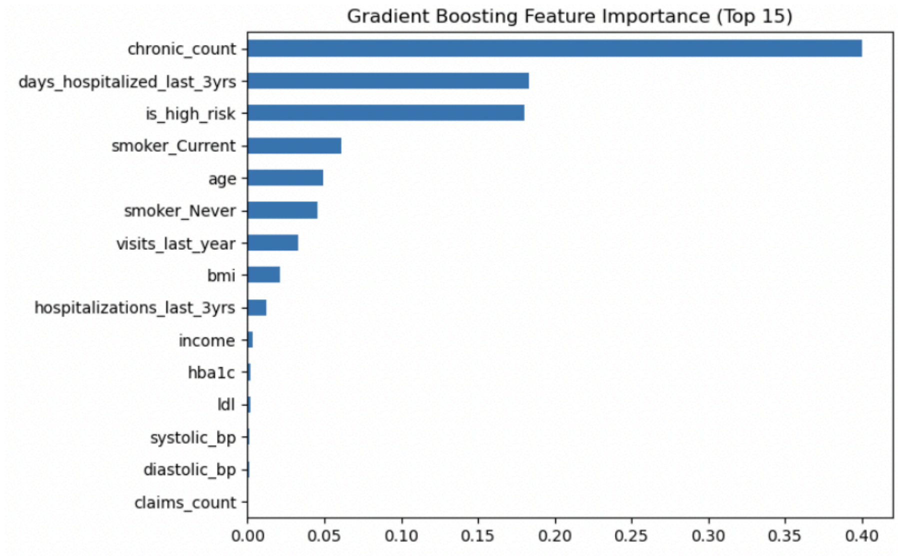
Results

All of our models, including Linear Regression, Bagging, Random Forest, and Gradient Boosting performed very similarly with an R^2 around 0.21. The specific results are as follows in the table below:

Model	**R^2**	**RMSE**
Linear Regression	0.213	0.752
Bagging	0.196	0.760
Random Forest	0.212	0.752
Gradient Boosting	0.216	0.751



The cross-validation results compare model performance using mean R2 across five folds. Gradient Boosting achieves the highest avg. R2 but all models perform similarly (~0.21). The small error bars (standard deviations ranging ~0.004-0.006) indicate that performance is highly stable across different data splits.



For gradient boosting, the number of chronic conditions diagnosed appeared to be the most significant feature for annual healthcare costs, with smoking status, age and hospital utilization also contributing meaningfully to predictions.

Discussion

Overall, our results show that the models all performed similarly, given the similar R squared and RMSE values. While Gradient Boosting did achieve the highest R squared, it was only slightly higher than the results for Linear Regression and Random Forest. The similarity in performance between the linear and non-linear methods suggests that the available predictors may only contain limited explanatory information regarding healthcare costs.

The lack of significant difference in model performance may be due to several reasons. The variables that we chose to analyze may not be strong predictors of medical costs, especially since we chose to exclude the cost variables. There may be key variables that were not included in the dataset but have a large impact on medical costs, such as additional health factors, diet, and a lack of lifestyle variables. To explore this idea of omitted variable bias further, we situate our project in the broader literature base, discussed below.

Our dataset did include specific disease-related information, including variables related to kidney disease, liver disease, a history of cancer, and the number of chronic illnesses. However, König et al. 2013 studies the effect of multimorbidity (the co-occurrence of 3+ chronic illnesses) on healthcare costs, and identifies two specific conditions as having the most influence on healthcare cost: Parkinson's disease and cardiac insufficiency. As these diseases are not specified within our dataset, it may be understood that the omission of specific disease variables prevents our models from successfully predicting healthcare costs. As another example, Frees et al. 2013 noted that self-rated mental health and self-rated physical health were very effective for linear regression cost prediction models, which our data again did not include.

Other work in the literature base supports our findings that cost predictors are most effective in predicting future costs, such as Duncan et al. 2016, and underscores our challenging decision to exclude it for the sake of usefulness, as we wanted our models to be helpful towards people which had not yet incurred large direct healthcare expenses.

However, it is worth noting that healthcare prediction literature consistently demonstrates the success of supervised learning techniques, especially boosted trees (Duncan et al. 2016), over linear regression so future work should more deeply examine why our models had such similar performances. Because our dataset was large, it made it difficult to try permutations of different parameters, which may have influenced the ability for different models to outperform one another. Additionally, our data was collected from multiple sources and stitched together rather than being derived from one cumulative survey, so future work should first identify trends in the literature base for how to best manage non-organic data.

Conclusion

In conclusion, our project demonstrates that predicting annual medical costs is a multifaceted problem that is influenced by various interconnected factors. By comparing Linear Regression, Bagging, Random Forest ,and Gradient Boosting models, we found that all the models achieved modest predictive power. However, the results still provide valuable insight into which patient characteristics are most associated with increased healthcare spending.

Our results show that factors such as the number of chronic conditions someone has been diagnosed with, the number of days they have been hospitalized in the past year, and whether or not someone is a smoker plays an important role in determining medical costs. This means that there are preventable factors that individuals can focus on to reduce their costs, such as choosing not to smoke, and consistently monitoring their health. Additionally, this tool can be helpful for people who have been recently diagnosed with a condition and want to understand how their annual medical costs or insurance plans might change.

Future work could improve predictive performance by incorporating more detailed medical history, behavioral information and additional lifestyle factors. Additional hyperparameter tuning and alternative ensemble methods may also help improve model accuracy. Overall, this project highlights the usefulness and limitations of applying machine learning methods to real-world healthcare data.

References

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